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Student Activity Pack

# How Do Wildfires Reshape Ecosystems?

Modeling Fire Ecology, Ecological Succession, and Climate-Fire Interactions

Grade 12L2 | NGSS HS-LS2-2 | Stability and Change

STUDENT NAME _____ DATE _____

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# Table of Contents

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1 Pre-Assessment ..... 3

2 Scoring Rubric ..... 4

3 Component Analysis ..... 5

4 Model Recording Page ..... 6

5 Simulation Observations ..... 7

6 Research & Extend ..... 8

7 Reflection Questions ..... 9

8 STEM Challenge ..... 10

# 1 Pre-Assessment

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**What Do You Know?** Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1. **Is wildfire always bad for the environment, or can it be beneficial? What do you think happened to forests before humans started fighting fires?**

2. **Why do you think wildfires have been getting larger and more destructive in recent decades? What factors contribute to this trend?**

3. **What is a prescribed burn and why would scientists intentionally set a forest on fire? What would be the purpose?**

4. **If you lived in a fire-prone area, what steps could protect your home and community from wildfire?**

# 2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

| Skill | Expert (4) | Proficient (3) | Developing (2) | Beginning (1) |
|---|---|---|---|---|
| **Identifying Components** | EXPERT<br>Correctly identifies and sorts ALL components as internal or external with detailed reasoning | PROFICIENT<br>Correctly identifies and sorts MOST components with some reasoning | DEVELOPING<br>Identifies components but has difficulty sorting or explaining | BEGINNING<br>Needs significant help identifying components |
| **Establishing Relationships** | EXPERT<br>Accurately describes ALL positive and negative relationships with clear evidence | PROFICIENT<br>Accurately describes MOST relationships with some evidence | DEVELOPING<br>Identifies some relationships but struggles with direction | BEGINNING<br>Needs help understanding how components connect |
| **Running Simulations** | EXPERT<br>Makes accurate predictions AND explains surprising results using evidence | PROFICIENT<br>Makes reasonable predictions and records clear observations | DEVELOPING<br>Runs simulations but predictions or observations are incomplete | BEGINNING<br>Needs help running simulations or recording data |
| **Explaining Results** | EXPERT<br>Uses multiple pieces of evidence to explain HOW and WHY systems interact | PROFICIENT<br>Uses evidence to explain system interactions | DEVELOPING<br>Gives partial explanations with limited evidence | BEGINNING<br>Needs help connecting evidence to explanations |

# 3 Component Analysis

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### What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

### Sort These Components

- Climate and Weather Conditions
- Fuel Load Accumulation
- Fire Intensity
- Soil Health and Seed Bank
- Post-Fire Vegetation Recovery
- Forest Carbon Balance
- Ecosystem Resilience

| EXTERNAL | INTERNAL |
|---|---|
| | |

### Explain your reasoning: Why did you sort each component the way you did?

# 4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

**? Explain ONE relationship in your model. What happens when one component changes?**

# 5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

## 1 Scenario 1: Historical Natural Fire

Fuel Load: Moderate (Natural) | Climate: Historical Average | Fire Type: Surface

MY PREDICTION

WHAT I OBSERVED

## 2 Scenario 2: Suppression Legacy Fire

Fuel Load: Maximum (100yr Accumulation) | Climate: Current | Fire Type: Mixed Surface/Crown

MY PREDICTION

WHAT I OBSERVED

## 3 Scenario 3: Climate Megafire

Fuel Load: Maximum | Climate: Extreme (Drought + Heat + Wind) | Fire Type: Crown

MY PREDICTION

WHAT I OBSERVED

# 6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

## Possible New Components

**Wildland-Urban Interface Density** — The number of homes and structures built in fire-prone areas at the boundary between developed and natural land — the WUI has expanded dramatically as housing development pushes into fire-adapted landscapes, putting more people and property at risk while making fire suppression politically mandatory even when ecologically harmful

**Atmospheric Smoke and Air Quality** — The impact of wildfire smoke on regional air quality and human health — megafires produce smoke plumes that travel thousands of miles, causing respiratory illness, cardiovascular problems, and premature death far from the fire itself, with the health burden falling disproportionately on outdoor workers and low-income communities

**Post-Fire Hydrology** — Changes to watershed behavior after fire — burned forests lose their ability to absorb rainfall, causing increased flooding, erosion, and sediment loading in downstream waterways, with high-intensity fires creating hydrophobic soil layers that dramatically increase runoff and debris flow risk for 3-10 years after the fire

## My Extension

**A Which new component would you add to the model?**

**B How would it connect to the existing components? (Draw or describe)**

**C What evidence or reasoning supports your connection?**

# 7 Reflection Questions

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### The Big Question

**For millions of years, fire was an essential force that maintained healthy ecosystems — so how did something natural become one of the most devastating consequences of climate change, and what does the new era of megafires mean for forests, communities, and the atmosphere?** Use evidence from your simulation to support your answers.

1. **How does your model demonstrate the paradox that preventing fire (suppression) actually increases fire risk? Trace the feedback loop from suppression to fuel accumulation to catastrophic fire.**

2. **What is the difference between fire in an ecosystem that evolved with it versus fire in a system with a century of accumulated fuel? How does your model show this difference in ecological outcomes?**

3. **Climate change is making fire seasons longer, hotter, and drier. How does your model show the interaction between climate amplification and the fire suppression legacy?**

4. **What does type-conversion mean in the context of post-fire recovery, and why is it ecologically significant? How does your model identify the conditions that lead to it?**

5. **Why is it politically difficult to implement prescribed burns and allow natural fire even when the science clearly supports it? How would you make the case to a community?**

# 8 STEM Challenge: Design a Community Wildfire Resilience Plan

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## YOUR ENGINEERING MISSION

Design a comprehensive wildfire management plan for a community at the wildland-urban interface that balances fire suppression for community protection with ecological fire management for long-term forest health.

### THE SCENARIO

A mountain community of 25,000 people is surrounded by fire-suppressed forest with extreme fuel loads. The region has experienced increasing drought and temperature, and fire season has lengthened by 50 days over the past 30 years. Two major fires in the past five years burned over 100,000 acres each and destroyed 500 homes. Ecologists warn that continued fire suppression will only make future fires worse. Your team must design a plan that protects the community while restoring natural fire regimes to surrounding forests — reducing the risk of catastrophic fire through proactive management rather than reactive suppression.

### Engineering Design Thinking

1. WHAT SPECIFIC FUEL REDUCTION STRATEGIES DOES YOUR PLAN IMPLEMENT — PRESCRIBED BURNS, MECHANICAL THINNING, OR BOTH — AND WHAT IS THE TREATMENT AREA AND TIMELINE?

2. HOW DOES YOUR PLAN CREATE DEFENSIBLE SPACE AROUND THE COMMUNITY WHILE ALLOWING ECOLOGICAL FIRE MANAGEMENT IN SURROUNDING FORESTS?

3. WHAT EARLY WARNING AND EVACUATION SYSTEMS DOES YOUR PLAN INCLUDE FOR COMMUNITY PROTECTION DURING INEVITABLE WILDFIRE EVENTS?

4. HOW DOES YOUR PLAN BALANCE SHORT-TERM POLITICAL PRESSURE TO SUPPRESS ALL FIRES WITH THE SCIENTIFIC EVIDENCE THAT FIRE SUPPRESSION INCREASES LONG-TERM RISK?

### My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

**! Explain your design using scientific evidence from the simulation:**

### Real Career Connection

Fire ecologists who study how fire shapes ecosystems and design prescribed burn programs work for the Forest Service, National Park Service, and land management agencies, earning $55,000-$110,000/year. Wildfire management specialists who coordinate fire suppression, community protection, and resource deployment earn $60,000-$120,000/year. Climate adaptation planners who integrate fire risk into community planning and land use policy earn $65,000-$125,000/year.